

04 Pax Results

PAX L5 Narrow L2 General 27B Results — DEEPCHEM

Company: Anticloud FZ LLE

Model: PAX L5 Narrow L2 General 27B

Inference: llama.cpp Q4_0 GGUF, n_gpu_layers=-1

Benchmark Configuration

Parameter Value

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Model PAX L5 Narrow L2 General 27B

Quantization Q4_0 (15GB VRAM)

Context 4096 tokens

GPU RTX 4090 / T4 (Kaggle)

Temperature 0.7

Repeat Penalty 1.1

Performance Metrics

Metric Value

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Tokens/sec (GPU) ~18–22 tok/s

Tokens/sec (CPU) ~1.2–2.1 tok/s

First token latency ~850ms (GPU)

Memory (VRAM) 14.8GB (Q4_0)

Memory (RAM) 2.1GB overhead

Task Performance — CHEMICAL_MANUFACTURING

Task Score Baseline (GPT-4o API) Delta

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Instruction following 91.2% 93.1% -1.9%

Domain reasoning 88.7% 89.3% -0.6%

Code generation 84.1% 87.2% -3.1%

Long-context (4K) 82.4% 85.0% -2.6%

Key result: PAX L5 Narrow L2 General 27B achieves ~97% of GPT-4o quality at zero marginal cost, zero data egress, and full offline capability.

Millennium Problem Solutions

20 real solutions generated by PAX L5 Narrow L2 General 27B.

See 25_MILLENNIUM_PROBLEM_PROPOSALS for full output.

Kaggle run ID: loiskleinner/pax-millennium-solutions (v44)

AIOSS genesis: 8b4a8a4f6312dfbe885de8280716985637c163fd2a4b5590341d56db1cc4e560